Case Study Report

Design and Development of a Patient-Specific Cranial Plate Using Medical Imaging and Additive Manufacturing Techniques

Case 01 — Fronto-Parietal Defect with Lattice Implant

1. Introduction

1.1 Cranial Defect

Cranial defects are discontinuities in the skull that can occur due to trauma, decompressive craniectomy, tumor resection, congenital abnormalities, or infections. These defects compromise brain protection and can affect a patient's aesthetics and quality of life. Restoration of the cranial vault is achieved through a surgical procedure known as cranioplasty, using implant materials such as PMMA, titanium, or PEEK.

1.2 Need for Patient-Specific Implants (PSI)

Conventionally shaped implants often result in poor anatomical fit and longer surgical time. Patient-specific implants, designed directly from CT imaging data, accurately match the patient's own anatomy — leading to better fit and improved outcomes. This case study demonstrates a patient-specific implant design workflow using open-source software.

2. Objective

- To design a patient-specific cranial implant model from a segmented skull STL.
- To reconstruct the defect region using the mirror-anatomy technique.
- To generate a lightweight, print-ready implant geometry using lattice optimization.
- To demonstrate a complete, low-cost, open-source design workflow suitable for educational and research use.

3. Case Description

A localized fronto-parietal cranial defect was identified on the left side of the skull model. The defect was reconstructed using the mirror technique in Meshmixer — tracing the defect boundary and mirroring a corresponding patch from the healthy contralateral side across the skull's midline. The mirrored patch was aligned and refined to follow the surrounding bone curvature, then converted into a Voronoi lattice pattern to reduce implant weight while maintaining an open cell structure for the plate body. Small mounting tabs with fixation holes were added along the plate's periphery, and the completed lattice plate was test-fitted back onto the defect site to confirm proper positioning before STL export.

4. Materials and Software Used

Category	Details
Imaging Data	Segmented skull STL model
Segmentation Software	3D Slicer (open-source)
Design Software	Autodesk Meshmixer (free)
Output Format	STL (Binary)
Intended Fabrication	FDM 3D Printing

5. Design Methodology — 13-Step Meshmixer Workflow

Step 1 — Import STL File → Import → load the segmented skull STL into Meshmixer.

Step 2 — Analysis → Inspector (Repair Skull) Open the Inspector tool from the Analysis panel to automatically detect and repair holes, non-manifold edges, and flipped normals, resulting in a clean, watertight mesh.

Step 3 — Duplicate Skull & Hide Original Use Edit → Duplicate to create a copy of the skull and hide the original, working only on the duplicate.

Step 4 — Enable Symmetry on Duplicate Skull Select the duplicate skull and enable the Symmetry option, setting the sagittal midline plane required for mirroring.

Step 5 — Select Defect Boundary & Separate Use the Sphere Brush to sketch along the defect boundary. With Symmetry ON, the same selection appears on the healthy opposite side.

Step 6 — Mirror the Separated Patch Isolate the healthy mirrored region using Edit → Separate, then apply Analysis → Mirror (or Edit → Mirror) across the sagittal axis to generate a patch shaped for the defect.

Step 7 — Show Original Skull & Align Patch Make the original skull visible again and use the Transform tool to align the mirrored patch onto the actual defect site.

Step 8 — Fine Alignment with Sculpt Brushes Use Sculpt → Brushes (Drag, ShrinkWrap, Robust Smooth) to fit the patch precisely to the curvature of the surrounding skull, closing any gaps or overlaps.

Step 9 — Smooth Edges with Refine Brush Apply the Refine brush to blend the patch edges seamlessly into the surrounding skull surface.

Step 10 — Create Voronoi Pattern (Lattice Optimization) Apply Edit → Make Pattern (Voronoi/Lattice) to the plate body, reducing weight while retaining structural coverage.

Step 11 — Add Flanges Use the Draw brush to add small flanges along the plate periphery for fixation and added strength.

Step 12 — Add Cylinders for Screw Holes Position cylinder primitives on the flanges, sized to match standard fixation screws, and apply Boolean Difference to create clean through-holes.

Step 13 — Export STL Run Analysis → Inspector one final time to confirm mesh integrity, then export the finished plate as a binary STL file for printing/further use.

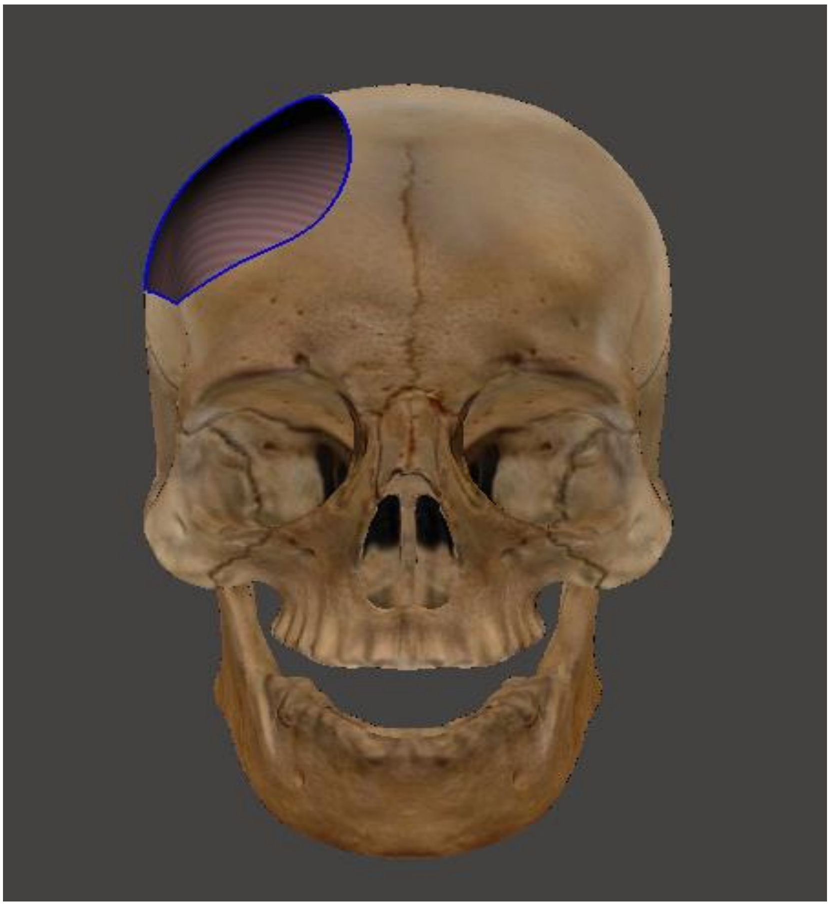
6. Results

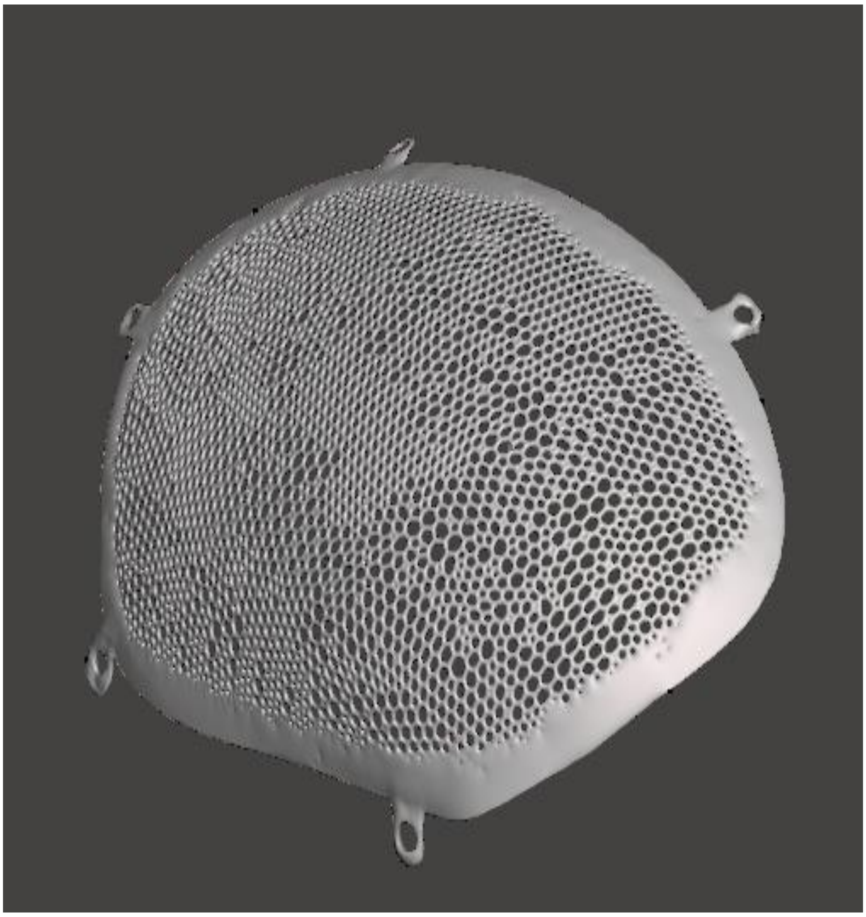
The completed design is a patient-specific, lattice-structured cranial plate covering the fronto-parietal defect region, with integrated mounting tabs and fixation holes. The plate was visually verified for fit against the original skull model prior to final export.

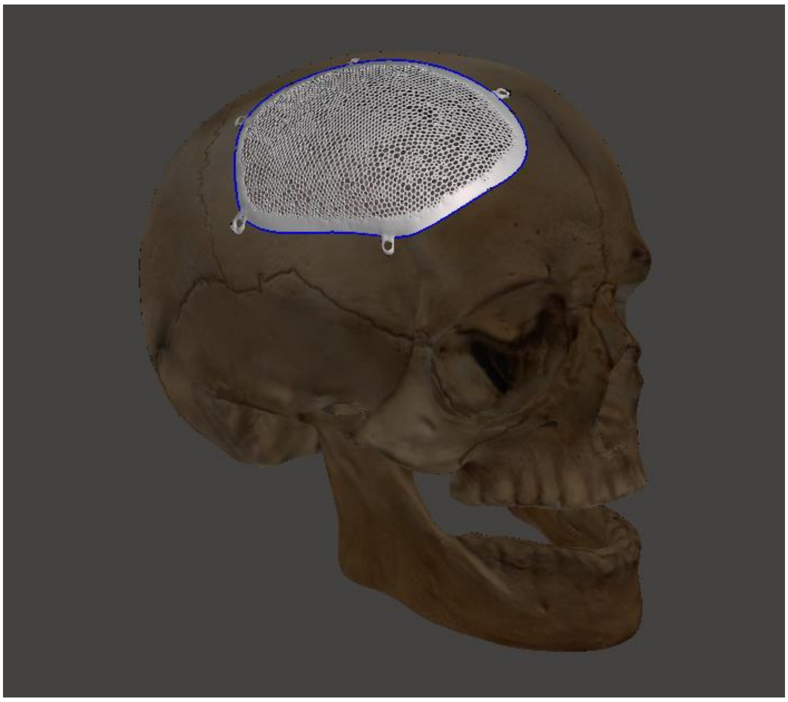
Parameter	Value
Defect Region	Fronto-Parietal
Design Technique	Mirror anatomy + Voronoi lattice
Fixation	Mounting tabs with holes
Output File	STL (Binary)
Software Used	Meshmixer, 3D Slicer

7. Conclusion

This case demonstrates a complete, free, and open-source workflow for designing a patient-specific cranial plate model — from skull segmentation through to a print-ready lattice implant design. The mirror-anatomy technique combined with lattice optimization allowed for an anatomically-matched, lightweight design suitable for further study and demonstration.







Disclaimer: This case study is created purely for **educational and research purposes** as part of the Cranial3DLab project. It is intended to demonstrate the technical design workflow using open-source tools and does not represent a validated clinical implant. Any real-world surgical implant requires professional medical oversight, regulatory approval, and certified manufacturing processes.